

MEDINA PRINCE ABDULAZIZ INTL, Saudi Arabia

WMO: 404300

Lat: 24.553N

Lon: 39.705E

Elev: 656

StdP: 93.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.9	(c) 11.3	(d) -17.2	(e) 0.9	(f) 31.3	(g) -14.2	(h) 1.2	(i) 29.1	(j) 9.9	(k) 21.7	(l) 8.8	(m) 20.0	(n) 2.3	(o) 50	(p) 0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.5	(c) 45.2	(d) 18.8	(e) 44.3	(f) 18.5	(g) 43.8	(h) 18.3	(i) 22.3	(j) 37.2	(k) 21.2	(l) 37.7	(m) 20.3	(n) 37.7	(o) 4.1	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.8	28.0	16.1	12.4	26.6	14.9	11.5	26.0	68.9	39.2	64.5	38.6	61.3	39.0	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.2	(c) 7.3	(e) 6.6	(f) 46.8	(g) 2.0	(h) 0.8	(i) 5.2	(j) 47.3	(k) 4.0	(l) 47.8	(m) 2.9	(n) 48.2	(o) 1.5	(p) 48.7		
(a) 9.3	(b) 8.2	(c) 7.3	(e) 1.0	(f) 23.9	(g) 1.7	(h) 1.3	(i) -0.3	(j) 24.8	(k) -1.2	(l) 25.6	(m) -2.2	(n) 26.3	(o) -3.4	(p) 27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	18.6	21.0	24.3	28.7	32.9	36.1	36.5	37.4	35.4	30.4	24.3	20.1
	DBStd	7.11	2.96	3.08	3.12	2.72	2.38	1.75	2.18	1.93	1.89	2.17	2.80	2.69
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	59	34	9	1	0	0	0	0	0	0	0	1	14
	CDD10.0	6878	267	307	443	561	709	782	821	849	763	634	429	313
	CDD18.3	3896	42	83	186	311	451	532	563	591	513	376	179	69
	CDH23.3	62648	313	762	2093	4327	7510	9693	10158	10704	9079	5666	1842	501
CDH26.7	43132	52	225	922	2502	5156	7298	7681	8226	6696	3523	764	87	
Wind	WSAvg	3.1	2.9	3.2	3.4	3.4	3.2	3.3	3.6	3.4	2.9	2.6	2.9	2.8
Precipitation	PrecAvg	60	7	3	7	8	4	1	1	4	1	4	14	7
	PrecMax	127	45	33	41	44	41	5	18	36	7	42	59	40
	PrecMin	6	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	31	11	7	11	11	10	1	4	8	1	10	19	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	33.0	37.0	40.1	43.2	45.2	46.6	46.2	44.8	40.2	35.1	30.2
		MCWB	15.2	15.9	15.6	17.0	18.0	18.1	19.3	19.4	18.6	17.3	16.5	15.5
	2%	DB	28.0	31.0	34.8	38.2	42.1	44.1	45.2	45.2	43.6	39.1	33.8	28.9
		MCWB	14.8	15.6	15.8	16.7	17.6	17.9	18.8	19.1	18.2	16.9	16.1	15.9
	5%	DB	26.8	29.2	33.0	37.1	41.0	43.2	44.1	44.8	42.8	38.1	32.2	27.6
		MCWB	14.5	14.9	15.4	16.4	17.3	17.7	18.4	18.9	18.0	16.7	16.1	15.1
	10%	DB	25.0	27.9	31.2	35.8	39.9	42.8	43.0	43.8	41.8	37.1	31.0	26.1
		MCWB	13.8	14.5	14.8	16.3	17.2	17.6	18.2	18.7	17.7	16.4	15.8	14.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	18.9	20.0	20.7	21.0	21.7	23.0	24.6	21.8	20.6	19.8	19.1
		MCDB	24.5	28.1	30.0	31.8	35.0	40.8	37.9	36.4	35.4	32.6	28.2	25.3
	2%	WB	17.5	18.0	18.4	19.5	20.0	20.3	21.2	22.9	20.4	19.5	18.9	18.0
		MCDB	24.5	26.9	29.0	31.5	35.2	39.5	38.7	36.7	37.6	32.1	27.4	25.3
	5%	WB	16.4	17.0	17.4	18.6	19.4	19.5	20.2	21.7	19.6	18.6	18.1	17.1
		MCDB	24.0	26.2	28.7	31.4	35.0	39.3	40.0	37.9	38.2	32.4	27.1	25.2
	10%	WB	15.3	16.0	16.3	17.7	18.5	18.5	19.5	20.6	18.9	17.9	17.3	16.0
		MCDB	22.9	25.1	28.3	31.6	35.0	39.5	40.1	39.2	38.2	33.1	27.3	24.3
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.1	12.7	13.1	13.1	13.3	12.5	12.5	13.0	13.1	11.7	11.3
		MCDBR	13.0	13.4	14.0	14.1	13.9	13.6	12.9	13.0	13.0	13.6	13.1	12.3
		MCWBR	6.7	6.4	5.9	5.3	4.9	4.5	4.3	4.4	4.4	5.0	5.4	5.9
	5% WB	MCDBR	10.8	11.5	11.9	12.6	12.7	13.3	12.6	12.7	12.7	12.3	10.9	10.7
		MCWBR	6.5	6.6	6.3	5.4	4.7	5.1	4.9	5.2	4.7	4.9	4.9	5.6
Clear-Sky Solar Irradiance	taub	0.369	0.420	0.463	0.517	0.539	0.525	0.539	0.593	0.531	0.482	0.430	0.391	
	taud	2.302	2.114	1.977	1.874	1.828	1.861	1.862	1.743	1.855	1.970	2.114	2.221	
	Ebn at Noon	882	856	837	800	778	783	772	730	766	784	805	838	
	Edh at Noon	116	150	181	205	214	206	205	231	202	171	139	122	
All-Sky Solar Radiation	RadAvg	4.50	5.38	6.45	7.06	7.48	8.01	7.70	7.09	6.57	5.71	4.68	4.21	
	RadStd	0.25	0.22	0.26	0.26	0.34	0.16	0.21	0.24	0.23	0.18	0.21	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.42	N/A	-1.90	N/A	N/A	N/A	N/A	N/A	+134	+132
(47) Regional (1 neighbor)		+0.46	N/A	-0.86	N/A	N/A	N/A	N/A	N/A	+156	+153

Nomenclature: See separate page